neuro-analog: A Cross-Architecture Compiler Framework for Analog In-Memory Computing

The Analog Adoption Gap

Analog in-memory computing (AIMC) has been reported to show $10\times$–$10^3\times$ projected inference energy gains over digital baselines in small-scale prototypes, but existing benchmarks (aihwkit, NeuroSim, MNSIM) primarily target CNN/MLP backbones. Modern ML has moved on; generative, implicit, and structured sequence models dominate, yet cross-architecture studies of whether their *computational structure* maintains task performance under analog nonidealities remain scarce.

We propose **analog amenability** as a structural property: a multidimensional profile across mismatch tolerance, ADC sensitivity, and idealized cost ratios (Fig. 4), and a compiler framework that probes it across seven families and maps each to Ark, an open analog circuit compiler with mismatch-aware optimization.

Framework Pipeline

neuro-analog ingests trained PyTorch models into a domain-tagged graph IR, simulates inference under analog-native noise, and lowers all seven families to Ark for hardware-aware retraining.

Stage 1 – Extract: Parse PyTorch models into a graph IR of ~30 circuit primitives (MVM, RC integration, Gibbs step, state-dependent matmul, etc.), tagged `ANALOG`, `DIGITAL`, or `HYBRID`.

Stage 2 – Map: Annotate nodes with noise specs (RRAM/PCM mismatch, kT/C thermal, shot noise), abstracted as additive Gaussians with literature-derived variance parameters at crossbar-output level; detect D/A boundary insertions.

Stage 3 – Simulate: `analogize()` replaces `nn.Linear`/`nn.Conv` with noise-injecting equivalents for inference-only simulation, paired with architecture-specific solvers (RC, fixed-point, Gibbs, SSM).

Stage 4 – Sweep: Monte Carlo mismatch sweeps (50 trials, $\sigma \in [0, 0.15]$, where σ is relative std. dev. of per-crossbar weight perturbations) and ADC ablations (25 trials, 2–16 bits).

Stage 5 – Ark Export: All seven families export to Ark's Constrained Dynamical Graph; Ark's `OptCompiler` emits JAX circuit classes for simulation and gradient-based retraining. Transformer maps FFN only.

Seven-Architecture Pilot Suite

Family (exemplar)	Dynamics	Analog Wt. (%)	Primary Bottleneck
Neural ODE	Continuous ODE	100%	Solver-step accumulation
SSM (S4D-style)	LTI diagonal	74%	Sequential state update
EBM (Hopfield)	Gibbs / Sampling	100%	Iterative convergence
Normalizing Flow	Continuous ODE	100%	Solver-step accumulation
DEQ	Implicit fixed-pt	100%	Fixed-point sensitivity
Transformer (FFN)	Discrete layers	91%	Softmax / DMMul (digital)
Diffusion	Multi-step SDE	100%	Quantization accumulation

Pilots: 542–16k params; 8×8 MNIST and 2D toy densities. Analog Wt. fraction is an upper bound because pilots lack production ops (LayerNorm and embeddings). Transformer row counts FFN only.

Key Contributions

1. **Cross-Family Pilot Benchmark.** Seven exemplars under a unified noise simulator, including implicit and continuous-time families (DEQ, Neural ODE, EBM, Flow) under-represented in prior analog work.
2. **Empirically-Derived Four-Class Taxonomy.** *Solver-step accumulating*, *fixed-point sensitive*, *quantization-bottlenecked*, and *mixed-domain* (Transformer, SSM), linking structure to dominant degradation channel based on our pilot suite sweeps.
3. **Domain-Tagged Compiler IR.** ~30 primitives with analog/digital/hybrid tags and a one-call `analogize()`; D/A boundary detection and per-node FLOP breakdown drives the cost model.
4. **Architecture-Specific Ark Backends.** CDG translation passes for all seven families enable Ark's adjoint optimizer to do hardware-aware retraining; we contribute the exporting, not the optimizer.

Cross-Architecture Mismatch Tolerance

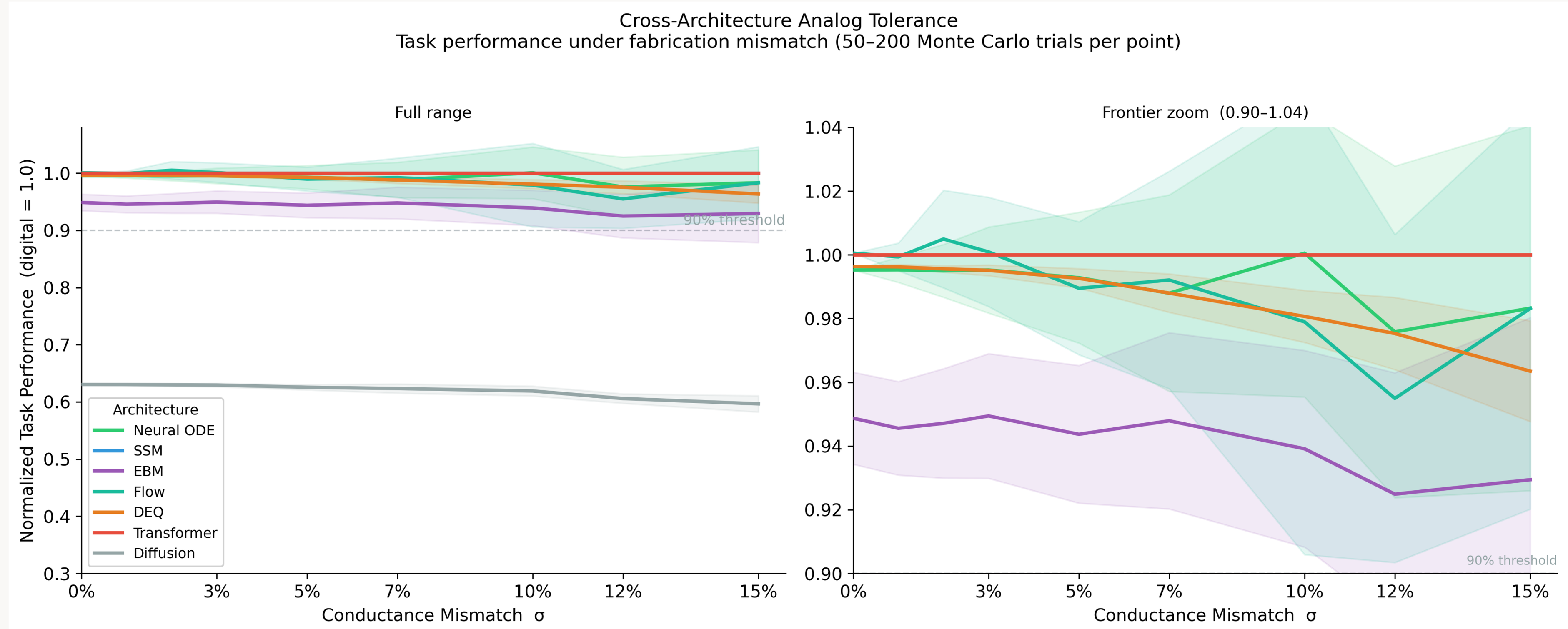


Fig. 1

Normalized quality = $Q(\sigma)/Q(0)$, where Q is accuracy (classifiers) or average log-likelihood (densities), relative to the $\sigma = 0$ analog baseline. 50 trials/point, $\sigma = 0$–0.15. **Caveat:** classification curves saturate; six of seven stay above 0.93 across the sweep. Continuous metrics are needed for fine-grained tiering.

Top tier: solver-step tolerant. Neural ODEs and Flows retain >95%; errors arise from integrating a fixed perturbed field, not from step-wise injected noise.

Middle tier: iterative but bounded. DEQ and EBM degrade earlier; DEQ trends downward, consistent with mismatch perturbing its fixed-point solution.

Outlier: Diffusion. Baseline quality is 0.63 and trends downward, suggesting the pilot is undertrained or structurally limited at this scale.

Practical context. Values are consistent with the literature: $\sigma \approx 0.03$–0.10 (RRAM) and up to 0.15 (PCM). Mismatch is sampled once per crossbar and held fixed (static-mismatch model).

ADC Precision Tradeoff

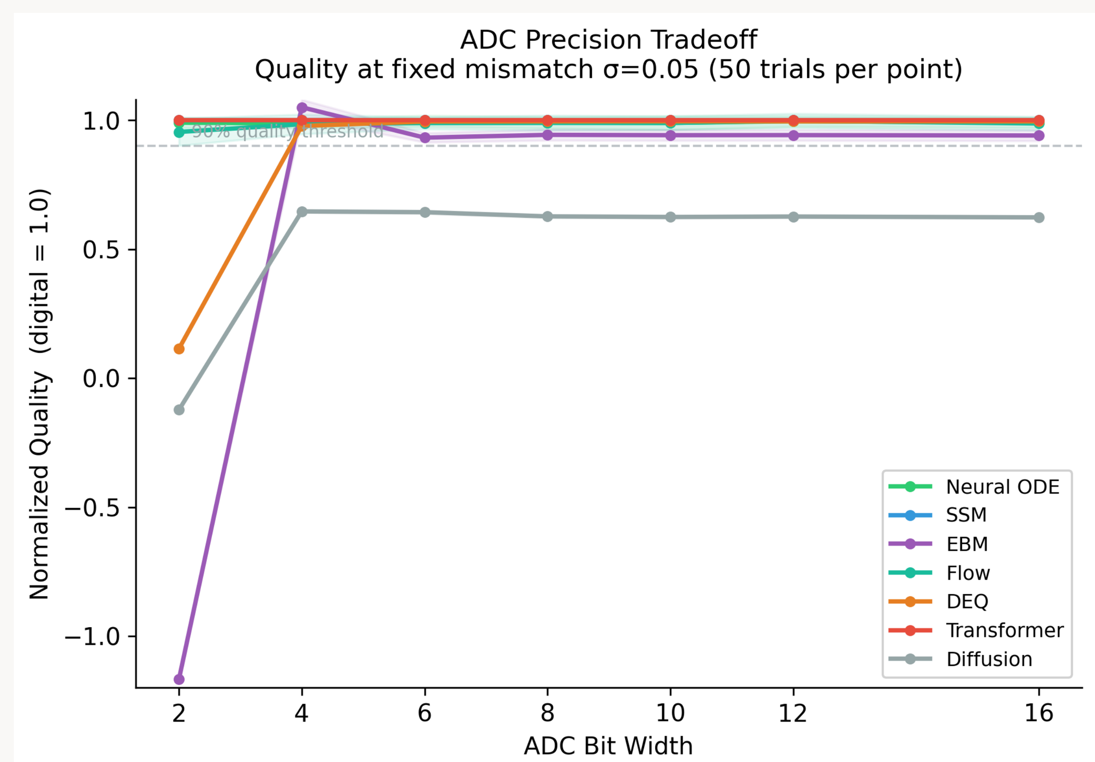


Fig. 2: Normalized quality (to a $\sigma = 0.05$, 16-bit ADC baseline) versus ADC resolution (2–16 bits, 25 trials/point) at $\sigma = 0.05$. Precision sensitivity is architecture-dependent.

DEQ 2-bit collapse. DEQ recovers at 4-bit ADC and is stable from 4 bits onward; coarse quantization of feedback is the dominant failure mode.

EBM 2-bit signed-metric anomaly. Quality is $\log p_{\text{analog}} - \log p_{\text{digital}}$ (negative means worse than digital); aggressive quantization pushes the analog mean below baseline. At 4 bits it is statistically iso-accurate, a metric artifact rather than unbounded degradation.

Diffusion floor. Diffusion stays at ~0.6 from 4–16 bits and never recovers to baseline; consistent with Fig. 1, training scale and architecture are the dominant bottlenecks, not ADC precision.

Idealized Cost Frontier & Scorecard

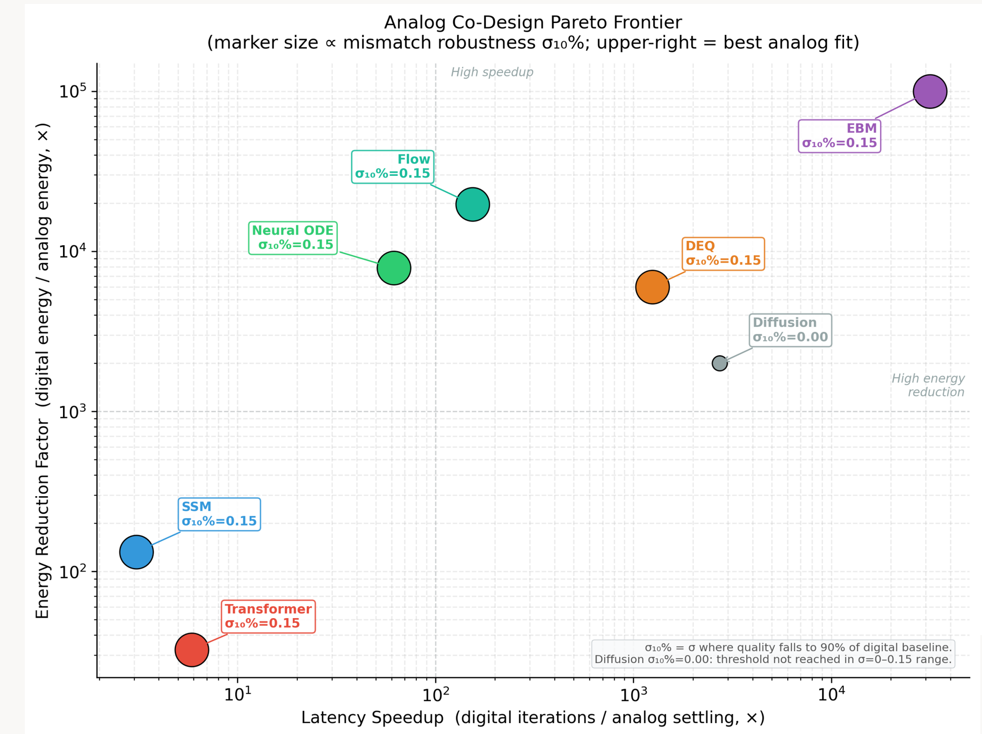
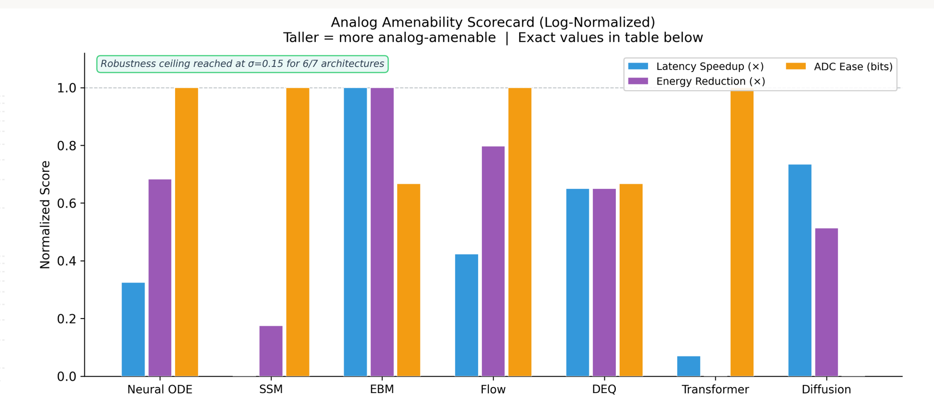


Fig. 3: Idealized latency / energy ratio.



Architecture	Speedup (×)	Energy Red. (×)	Min ADC (bits)
Neural ODE	61.7	7858.8	2
SSM	3.1	132.6	2
EBM	31509.6	99945.1	4
Flow	154.2	19647.1	2
DEQ	1249.8	5996.7	4
Transformer	5.9	32.5	2
Diffusion	2727.1	1997.9	16

Fig. 4: Normalized amenability scorecard.

Latency "speedup" and energy "reduction" are *idealized ratios of digital iteration count to analog settling iteration*, assuming per-iteration cost is comparable. These are upper-bound structural figures that exclude peripheral, ADC/DAC, control, and memory overhead, and they are not wall-clock measurements. We assume fully-parallel crossbar mapping. An optimized GPU baseline would reduce the effective gap.

Iterative families dominate the structural ratio. EBM (~31,000×), DEQ (~1,250×), and Diffusion (~2,700×) trade many digital iterations for one analog relaxation. These ratios assume a naive digital iteration count comparison, and ratios are large because iteration counts are high.

Non-iterative families are modest. Transformer FFN (~5.9×) and SSM (~3.1×) are bottlenecked by digital softmax, attention, and sequential state updates, all off-crossbar in our mapping.

- **Iterative families:** EBM, DEQ, and Diffusion have the largest structural gains in the cost frontier but require careful ADC budgeting per scorecard (Fig. 4).
- **Balanced candidates:** Neural ODEs and Flows show high mismatch tolerance, modest ADC needs, and meaningful structural gains.
- **Conservative middle:** Transformer and SSM have modest structural gains and a clean mapping onto existing tooling.

Implications for Hardware–Software Co-Design

- **Hardware architects:** A single global ADC spec is suboptimal because DEQs and Diffusion show much sharper 2-bit degradation than Neural ODEs and Flows; ADC precision should be a per-region tunable resource.
- **ML researchers:** Implicit models (DEQ, EBM) are structurally poised to gain the most when mapped as analog feedback / relaxation circuits, which is Ark's natural output, rather than as unrolled digital iterations.
- **Compiler designers:** Substrate-aware scheduling must treat D/A boundaries as a first-class cost; even FLOP-matched models with different ADC crossings behave as different programs from a hardware-cost perspective.

Status, Limitations & Next Steps

- **Scope:** All plots use pilot models (542–16k parameters) on 8×8 MNIST and 2D toy densities. Conclusions are preliminary and structural, not deployment-grade.
- **Honest gaps:** Transformer attention stays digital; Drift, write noise, IR drop, and stuck-at faults are not modelled, deferred to v2. `HardwareProfile` is literature-parameterized, not silicon-calibrated; cost model omits peripheral and dataflow terms.
- **Code:** `https://github.com/apumutyala/neuro-analog` – IR, simulator, mappers, seven Ark exporters, sweep harnesses, and pilot checkpoints.
- **Next steps:** Train CIFAR-10 / WikiText-2 suite; calibrate against published AIMC chip data; fix DEQ training (spectral normalization); switch to continuous metrics; extend extractors for more models.

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